

B.Tech. CSE (IV YEAR - VII SEM) (2026-2027)
**DEPARTMENT OF COMPUTER ENGINEERING &
APPLICATIONS**

GLA University



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Project Title: Multi Agents – Voice Based ERP System

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Project Synopsis: Multi Agents – Voice Based ERP System

0. Cover

- **Project title:** Multi Agents – Voice Based ERP System
- **Team name & ID:** Team Pixelate
- **Institute / Course:** GLA UNIVERSITY, Semester-Project
- **Version:** v1.0
- **Date:** 9 Sep 2026

Revision history

Version	Date	Author	Change
v0.1	9 Sep 2026	Arjun Singh Rajput	Initial draft
v1.0	9 Sep 2026	Team Pixelate	Complete synopsis with technical specifications

1. PROBLEM STATEMENT

- Modern Enterprise Resource Planning (ERP) systems integrate different business functions such as finance and accounting, sales, purchase, inventory, customer management and other operational activities into a centralized system. However, although ERP systems collect a large amount of structured business data, extracting meaningful insights from this data often requires users to navigate multiple modules, reports and dashboards.
- Traditional ERP systems mainly depend on forms, menus, tables, predefined reports and manually generated queries. A business user who wants to ask a simple question such as “What is the current financial position of the business?”, “Which products are generating the highest revenue?”, or “Which customers have increasing outstanding payments?” may need to manually access and combine information from different ERP modules.
- Another major challenge is that data generated by different modules is often analyzed independently. For example, sales data may be available in the sales module, purchase information in the purchase module, inventory information in the inventory module and financial transactions in the accounting module. Without effective cross-module analysis, important relationships and business patterns can remain hidden.

- Existing AI-based solutions can provide conversational interaction and automated analysis, but their responses may not always be directly connected to the organization's actual ERP data. This creates a risk of unsupported, inaccurate or non-verifiable AI-generated conclusions. For business decision-making, it is important that every important insight or recommendation can be traced back to relevant ERP records and supporting evidence.
 - Business users also need more than descriptive reports. They require predictive decision support, such as identifying possible cash-flow issues, detecting unusual transaction patterns, forecasting business trends and highlighting potential operational risks. At the same time, users should be able to understand why an AI system has produced a particular insight or recommendation.
 - Therefore, there is a need for an intelligent ERP framework that can combine natural language interaction, multi-agent AI, cross-module ERP analytics, evidence-grounded responses, explainable business intelligence and predictive decision support within a unified system.
 - The proposed project aims to address this problem by developing a voice-based intelligent ERP system in which users can interact with the ERP using natural language and voice commands. Instead of manually navigating through multiple screens and reports, users will be able to ask business-related questions conversationally and receive responses based on actual ERP data.
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2. PROPOSED PROJECT / APPROACH

The proposed system will be an **AI-enabled, voice-based ERP platform** designed to make business management and analysis more natural, intelligent and accessible. The core system will combine conventional ERP functionality with a **Multi-Agent Artificial Intelligence framework** capable of understanding user queries, identifying the required ERP information, performing analysis and generating evidence-backed responses.

2.1 Voice-Based Natural Language Interaction

The system will provide a conversational interface through which users can interact with the ERP using **voice or natural language queries**.

For example, a user may ask:

- *“Show today's sales.”*
- *“What is our total outstanding amount?”*
- *“Which products had the highest sales this month?”*
- *“Compare this month's sales with last month.”*
- *“Which customers have overdue payments?”*

The system will interpret the user's intent and convert the request into appropriate ERP operations or analytical queries. This reduces dependency on manually navigating complex ERP menus and reports.

2.2 Multi-Agent AI Architecture

Instead of depending on a single AI component for every task, the proposed framework will use multiple specialized AI agents. Each agent will be responsible for a specific category of ERP-related tasks.

Possible agents include:

- **Query/Orchestrator Agent** – understands the user's request and coordinates the required agents.
- **Finance Agent** – handles accounting and financial analysis.
- **Sales Agent** – analyzes sales transactions, customers and revenue-related information.
- **Purchase Agent** – analyzes suppliers, purchases and procurement-related information.
- **Inventory Agent** – analyzes stock levels, movement and inventory-related patterns.
- **Analytics Agent** – combines information from multiple modules to generate cross-module insights.
- **Prediction/Decision Support Agent** – identifies trends and provides data-driven forecasts or recommendations.
- **Evidence/Validation Agent** – verifies that generated responses are supported by relevant ERP records.

The agents will work together rather than operating as isolated modules. This will allow the system to answer questions that require information from multiple ERP functions.

2.3 Evidence-Grounded ERP Intelligence

A key component of the proposed system is **evidence grounding**. AI-generated responses should be based on actual information available in the ERP rather than unsupported assumptions.

When a user asks a question, the system will identify the relevant ERP records, reports or calculated results and use them as evidence for generating the response.

For example, instead of simply stating:

“Sales are expected to increase next month.”the system should provide an explanation based on relevant historical sales data and indicate the factors that contributed to the prediction.

This approach will improve the **reliability, traceability and usefulness** of AI-generated business insights.

2.4 Cross-Module ERP Analytics

The proposed framework will enable analysis across different ERP modules.

For example, a query about profitability may require information from:

Sales → Purchase → Inventory → Accounting

The AI framework will combine relevant information from these modules and generate a consolidated insight.

This cross-module approach can help users identify relationships such as:

- Sales performance versus inventory availability
- Revenue versus purchase costs
- Customer sales versus outstanding payments
- Inventory movement versus sales demand
- Purchase trends versus profitability
- Financial transactions versus operational performance

Thus, the system will move beyond isolated module-level reporting towards **integrated business intelligence**.

3. Team and Roles

Member	Role	Roll No.	Availability	Contact
Arjun Singh Rajput	Project Lead – Agentic AI & Research	2315800015	10 hrs/wk	arjun.rajpud_cs.h23@gla.ac.in
Keshav Agrawal	NLP/LLM, ML & Evaluation Lead	2315800043	8 hrs/wk	keshav.agrawal_cs.h23@gla.ac.in
Jatin Khetan	Frontend & UX Lead	2315800039	8 hrs/wk	jatin.khetan_cs.h23@gla.ac.in
Somesh Rajput	Backend & Database Lead	2315800080	8 hrs/wk	somesh.rajpud_cs.h23@gla.ac.in

4. Expected Outcomes

The proposed project is expected to produce a functional **voice-based intelligent ERP system** with an integrated multi-agent AI framework.

The major expected outcomes are:

4.1 Voice-Based ERP Interaction

Users will be able to interact with the ERP using natural language and voice commands instead of depending completely on traditional menus, forms and manually generated reports.

4.2 Automated Cross-Module Analysis

The system will be capable of retrieving and combining relevant information from multiple ERP modules to answer complex business questions.

For example, a single query can potentially combine sales, inventory and accounting information to generate a consolidated business insight.

4.3 Evidence-Grounded AI Responses

AI-generated responses will be connected to actual ERP data and supporting evidence. This will improve the reliability and traceability of generated information and reduce the possibility of unsupported responses.

4.4 Explainable Business Intelligence

The system will provide understandable explanations along with analytical results wherever applicable, helping users understand **how and why** a particular result or recommendation was generated.

4.5 Predictive Decision Support

The system will use historical ERP information to identify trends and generate predictive insights that can assist users in business planning and decision-making.

4.6 Multi-Agent Intelligent Processing

Specialized AI agents will divide ERP-related tasks according to their domain and collaborate through an orchestration mechanism. This will make the framework more modular, scalable and suitable for handling complex ERP queries.

4.7 Improved User Experience

The conversational interface is expected to reduce the effort required to locate information within the ERP. Users will be able to ask questions in a more natural manner instead of learning complex report structures or navigation workflows.

4.8 Integrated ERP and AI Platform

The final outcome will be a unified platform combining:

ERP Operations + Voice Interaction + Multi-Agent AI + Cross-Module Analytics + Explainability + Predictive Decision Support

The project will therefore demonstrate how Artificial Intelligence can be integrated with an ERP environment to transform traditional business software from a **record-management system into an intelligent, conversational and decision-support platform.**